

SAFETY DATA SHEET

Acetone (Reagent Grade)

SECTION 1: IDENTIFICATION

Product Name: Acetone (Reagent Grade)

Product Code: ACE-RG-4L

CAS Number: 67-64-1

Manufacturer: LabChem Industries

Address: 567 Research Park Dr, Raleigh, NC 27606

Phone: (919) 555-6789

Emergency Phone: 1-800-535-5053 (Infotrac)

Recommended Use: Solvent for cleaning, degreasing, chemical synthesis

Restrictions on Use: Not for food or drug use

SDS Version: 5.0

Revision Date: 2023-09-20

SECTION 2: HAZARD(S) IDENTIFICATION

Signal Word: Danger

GHS Classification: Flammable Liquid Category 2

H225 Highly flammable liquid and vapor

H319 Causes serious eye irritation

H336 May cause drowsiness or dizziness

P210 Keep away from heat, hot surfaces, sparks, open flames

P233 Keep container tightly closed

P240 Ground and bond container and receiving equipment

P241 Use explosion-proof electrical, ventilating, lighting equipment

Other Hazards: Prolonged exposure may cause CNS depression.

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

Acetone 67-64-1 99.5%

Water 7732-18-5 0.5%

SECTION 4: FIRST-AID MEASURES

Inhalation: Remove person to fresh air. If not breathing, give artificial respiration.

Skin Contact: Wash with plenty of soap and water.

Eye Contact: Flush eyes with water for at least 15 minutes.

Ingestion: Do NOT induce vomiting. Rinse mouth.

Notes to Physician: No specific antidote. Treat symptomatically.

SECTION 5: FIRE-FIGHTING MEASURES

Suitable Extinguishing Media: Dry chemical, CO2, alcohol-resistant foam, water spray.

Special Hazards: Vapors may travel to source of ignition and flash back.

Protective Equipment: Full protective clothing and SCBA for firefighters.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions: Remove all ignition sources. Ventilate area.

Environmental Precautions: Do not allow to enter drains or waterways.

Containment/Cleanup: Absorb with inert material (vermiculite, sand).

SECTION 7: HANDLING AND STORAGE

Handling: Use with adequate ventilation. Keep away from heat, sparks, and open flame.

Storage: Store in cool, dry, well-ventilated area. Keep away from oxidizers.

Incompatible Materials: Strong oxidizers, strong acids, strong bases

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

TWA: 750 ppm (OSHA PEL)

STEL: 1000 ppm (ACGIH)

Engineering Controls: Mechanical ventilation or local exhaust.

Respiratory Protection: Organic vapor cartridge respirator.

Hand Protection: Nitrile or neoprene gloves.

Eye Protection: Safety glasses with side shields.

Skin Protection: Lab coat or chemical-resistant clothing.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Physical State: Liquid

Color: Colorless

Odor: Sweet, pungent, characteristic

pH: Neutral (7)

Melting Point: -94.7 C

Boiling Point: 56.05 C

Flash Point: -20 C (closed cup)

Evaporation Rate: 7.7 (Butyl Acetate = 1)

Upper Explosive Limit: 12.8%

Lower Explosive Limit: 2.5%

Vapor Pressure: 231 mmHg at 20 C

Vapor Density: 2.0 (Air = 1)

Relative Density: 0.791 g/cm³ at 20 C

Solubility: Miscible with water

Auto-ignition Temperature: 465 C

Viscosity: 0.32 mPa s at 20 C

Molecular Weight: 58.08 g/mol

Molecular Formula: C₃H₆O

SECTION 10: STABILITY AND REACTIVITY

Stability: Stable under recommended storage conditions.

Conditions to Avoid: Heat, flames, sparks, static discharge.

Incompatible Materials: Strong oxidizers, strong acids, strong bases.

Hazardous Decomposition Products: Carbon monoxide, carbon dioxide.

Possibility of Hazardous Reactions: Vapors may form explosive mixtures with air.

SECTION 11: TOXICOLOGICAL INFORMATION

Skin Corrosion/Irritation: Mild irritant. May cause drying and cracking of skin.

Eye Damage/Irritation: Causes serious eye irritation.

Carcinogenicity: Not classifiable (IARC Group 3).

LD50 (oral, rat): 5800 mg/kg

SECTION 12: ECOLOGICAL INFORMATION

Persistence/Degradability: Readily biodegradable.

Bioaccumulative Potential: Low potential (Log Kow = -0.24).

Mobility in Soil: High mobility due to high water solubility.

LC50 (Fish): 8300 mg/L (96h, fathead minnow)

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Method: Incinerate in approved facility. Follow local regulations.

EPA Waste Code: U002 (if discarded as a commercial chemical product)

Contaminated Packaging: Empty containers may retain residue.

SECTION 14: TRANSPORT INFORMATION

UN Number: UN1090

Proper Shipping Name: Acetone

Hazard Class: 3

Packing Group: II

DOT Description: UN1090, Acetone, 3, PG II

SECTION 15: REGULATORY INFORMATION

SARA 311/312: Fire Hazard, Immediate (Acute) Health Hazard

SARA 313: Not listed

CERCLA RQ: 5000 lbs (2270 kg)

RCRA Waste Code: U002

TSCA Status: Listed on TSCA Inventory

California Prop 65: Not listed

SECTION 16: OTHER INFORMATION

Revision Notes: Section 9 updated with additional physical properties.

Disclaimer: Information is based on current knowledge.